# ICB response prediction

## Nemotron clinical copilot

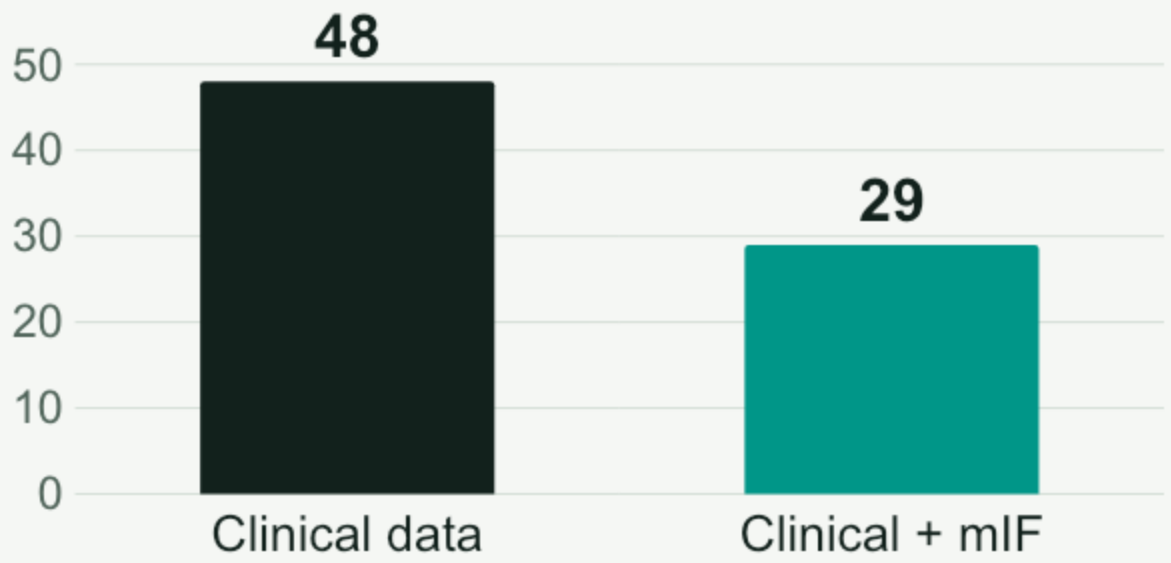
Architecture and model integration

Research demonstration



# Research scope

## Available patients




## Research question

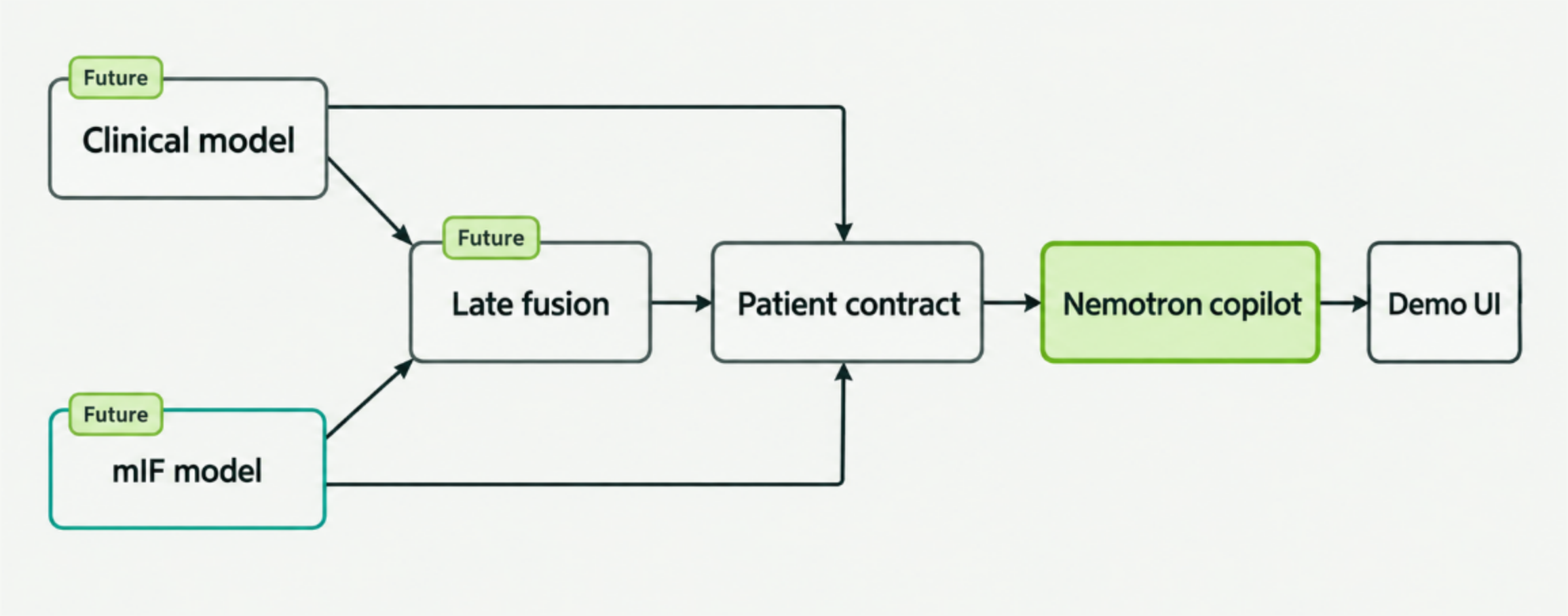
Does combining clinical and mIF information improve prediction of ICB responders?

## Planned comparison

Clinical-only, mIF-only and late-fusion predictions on the same evaluation patients.

The 29 paired patients are a subset of 48. H&E and extra mIF batches remain possible extensions.

# System architecture




Models export scores, features and context. This repository implements the copilot and its interfaces.

# Copilot workflow

| Step | What the component does |
| --- | --- |
| 1. Assemble and validate | Join by patient ID. Check scores, metadata and duplicate records. |
| 2. Preserve the facts | Report supplied scores and features. Keep missing modalities visible. |
| 3. Generate locally | Apply the chat template and request a constrained JSON interpretation. |
| 4. Screen the prose | Check format, numeric claims and unsupported clinical language. |
| 5. Use fallback when needed | Replace rejected model text with a deterministic evidence summary. |
| 6. Publish results | Preserve predictions, limitations, source, quality and provenance. |

Mock experiments and interfaces are implemented. Actual A30 generation still needs verification.

# Shared patient contract


```
{
  "schema_version": "1.0",
  "patient_id": "P12",
  "predictions": {
    "clinical": {"probability": 0.21},
    "mif": {"probability": 0.78},
    "fused": {"probability": 0.57}
  },
  "response_definition": "Not specified"
}
```


## Patient identity

One canonical patient ID and one prediction per patient per modality.

## Score metadata

Model version, calibration status and optional threshold travel with each score.

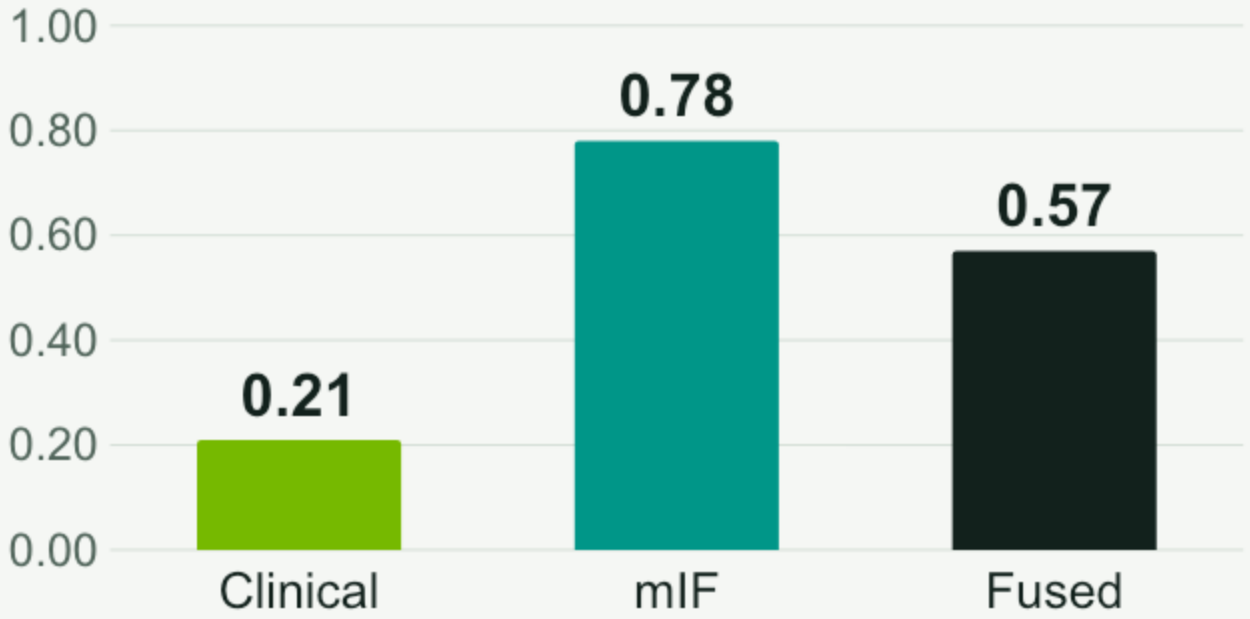
## Feature evidence

Measurement direction and signed attribution are separate.

Positive class: ICB responder. The team has not yet agreed the responder definition.

# Synthetic case: discordant predictions

## P12 / synthetic model scores




**“Clinical and spatial mIF predictions are discordant and warrant review of the supplied evidence.”**

Clinical–mIF gap: 0.57
Calibration: unknown
Human review: required

This is deterministic mock output. The supplied fused score does not establish better performance.

# One command on the HPC

```
bash run_hpc.sh
```

| Stage | Scheduled job behavior |
| --- | --- |
| SLURM allocation | Submit to red with one A30 GPU. Return the job ID and log paths. |
| Environment and checks | Create the isolated, pinned environment. Verify runtime and GPU. |
| Public model preparation | Download public weights on first setup. Reuse the local snapshot. |
| Patient inference | Use local weights with offline settings and telemetry disabled. |
| Experiment artifacts | Write narrative files, report, manifest and the metrics file. |

Network-limited nodes: --prepare-only first, then --offline. Submission returns before completion.

# Connections to future models

| Owner / workstream | Handoff to the copilot |
| --- | --- |
| Isis / clinical model | Clinical prediction CSV plus clinical features and context. |
| Alyssa / mIF model | mIF prediction CSV plus feature metadata and canonical patient IDs. |
| Fusion owner / TBD | Fused prediction CSV with model provenance for paired patients. |
| Paulo / copilot | Assemble, validate and interpret the combined patient records. |
| Junhao / demo UI | Read structured narrative results or call the local interpretation API. |

Future H&E predictions require a versioned contract and adapter change. Raw images stay upstream.

# Demo interfaces

## Completed experiment files

Junhao loads narratives.jsonl into the patient selector.

The UI shows scores, feature evidence, limitations and review status.

**Proposed starting point for the hackathon**

## Optional local service

GET /health
POST /v1/interpret
POST /v1/interpret-batch

A reused local model handles requests. The UI calls from the same node or through an approved tunnel.

Both modes expose the same structured result. The demo should keep source and limitations visible.

# Evaluation and review

### Prediction quality

Use the same eligible patients across modalities.

Agree the endpoint and preserve patient-level folds.

Inspect AUROC and Brier score on paired outputs.

### Narrative quality

Check reproduced scores and supplied features.

Review disagreement and missing modalities.

Inspect unsupported claims, source and limitations.

**A successful mock run establishes interface behavior. Clinical validity and multimodal benefit require further evidence.**

# Decisions needed for the real-data demo

Proposed starting choices and owners for team confirmation

| Decision | Suggested owner | Starting choice to confirm |
|---|---|---|
| Responder endpoint | Clinical owner TBD + Isis | Agree the responder definition and assessment window. |
| Patient identity | Isis + Alyssa | Use canonical IDs. Resolve slide aggregation upstream. |
| Evaluation design | Modeling team | Share patient-level folds. Start with the paired cohort. |
| Late-fusion method | Owner TBD | Assign an owner. Prespecify a simple fusion baseline. |
| Export metadata | All model owners | Fix class mapping, attribution scales and score metadata. |
| Demo and acceptance | Paulo + Junhao Reviewer TBD | Choose demo mode. Verify A30 output and review cases. |

H&E and extra mIF batches also need a scope decision before inclusion.

# Completion checkpoints

| Checkpoint | Evidence of completion |
| --- | --- |
| Contract agreement | The team records the endpoint, IDs, splits and model exports. |
| A30 smoke experiment | Paulo generates local Nemotron output from the synthetic cases. |
| Real prediction handoff | Model owners supply held-out clinical, mIF and fused scores. |
| Review and comparison | The team inspects paired metrics and representative narratives. |
| Demo integration | Junhao displays each patient’s scores, evidence and review status. |

Ready now: mock experiment, HPC runner and integration interfaces. Pending: real GPU and model handoffs.